# 1 Connect Jupyter to Mysql

[41]:
```
from sqlalchemy import create_engine
import pandas as pd
import matplotlib.pyplot as plt
username = "root"
password = "Santhosh%40578"
host = "localhost"
port = 3306
database = "ecommerce"
connection_url = f"mysql+pymysql://{username}:{password}@{host}:{port}/
↪{database}"
engine = create_engine(connection_url)
connection = engine.connect()
df = pd.read_sql("SHOW TABLES", con=connection)
print(df)
```

```
  Tables_in_ecommerce
0        dataanalysis
1   e-commerce dataset
2           ecommerce
3              orders
4               sales
5          sales_data
```

## 1.1 Run queries

[42]:
```
df=pd.read_sql('select * from sales_data',con=connection)
df
```

[42]:

| | id | name | buydate | productname | sales |
|---|---|---|---|---|---|
| 0 | 120 | santhosh | 2023-01-10 | Electronics | 40000 |
| 1 | 121 | vamsee | 2023-01-12 | clothing | 90000 |
| 2 | 122 | shanmukha | 2023-01-14 | books | 50000 |
| 3 | 123 | thiru | 2023-01-15 | sports | 39000 |
| 4 | 124 | shanmukha | 2023-01-18 | beauty products | 60000 |
| 5 | 125 | uday | 2023-01-20 | Electronics | 30000 |

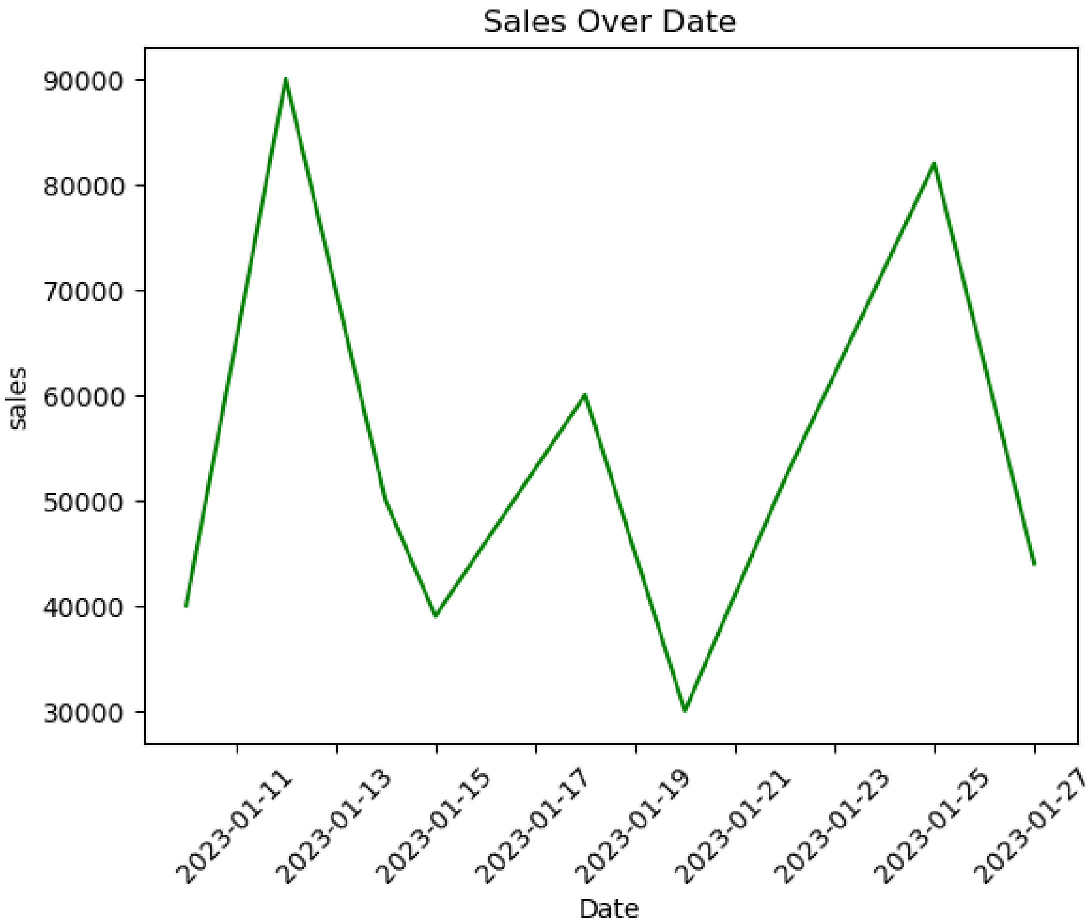
| | | | | | |
|---|---|---|---|---|---|
| 6 | 126 | sai | 2023-01-22 | sports | 52000 |
| 7 | 127 | vignesh | 2023-01-25 | beauty products | 82000 |
| 8 | 128 | varun | 2023-01-27 | clothing | 44000 |

[72]:
```
df=pd.read_sql('select buydate,sales from sales_data',con=connection)
df
```

[72]:

| | buydate | sales |
|---|---|---|
| 0 | 2023-01-10 | 40000 |
| 1 | 2023-01-12 | 90000 |
| 2 | 2023-01-14 | 50000 |
| 3 | 2023-01-15 | 39000 |
| 4 | 2023-01-18 | 60000 |
| 5 | 2023-01-20 | 30000 |
| 6 | 2023-01-22 | 52000 |
| 7 | 2023-01-25 | 82000 |
| 8 | 2023-01-27 | 44000 |

[80]:
```
df=pd.DataFrame(df)
plt.plot(df['buydate'],df['sales'],color='g')
plt.xticks(rotation=45)
plt.ylabel('sales')
plt.xlabel('Date')
plt.title('Sales Over Date')
plt.show()
```

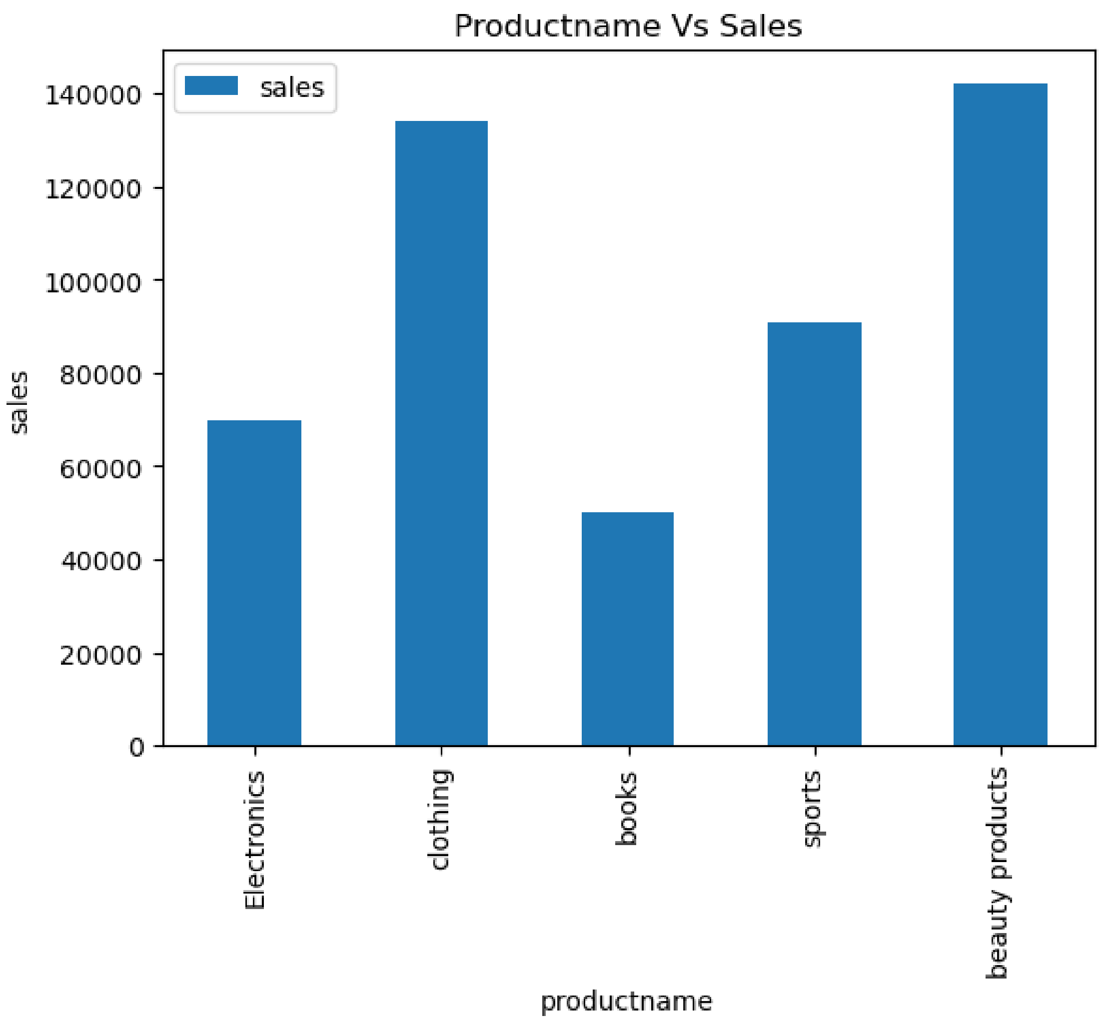



[43]:
```
df=pd.read_sql('select productname,sum(sales) as sales from sales_data Group By␣
↪productname',con=connection)
df
```

[43]:

| | productname | sales |
|---|---|---|
| 0 | Electronics | 70000.0 |
| 1 | clothing | 134000.0 |
| 2 | books | 50000.0 |
| 3 | sports | 91000.0 |
| 4 | beauty products | 142000.0 |

[50]:
```
df.plot(kind='bar',x='productname',y='sales')
plt.ylabel('sales')
plt.title('Productname Vs Sales')
plt.show()
```

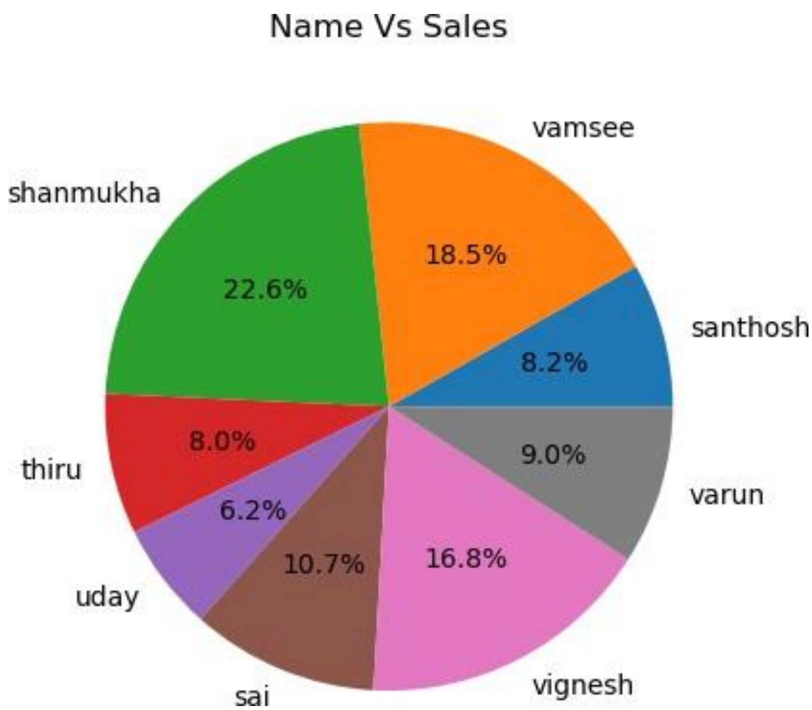



[56]:
```
df=pd.read_sql('select name,sum(sales) as sales from sales_data Group By␣
↪name',con=connection)
df
```

[56]:

| | name | sales |
|---|---|---|
| 0 | santhosh | 40000.0 |
| 1 | vamsee | 90000.0 |
| 2 | shanmukha | 110000.0 |
| 3 | thiru | 39000.0 |
| 4 | uday | 30000.0 |
| 5 | sai | 52000.0 |
| 6 | vignesh | 82000.0 |
| 7 | varun | 44000.0 |

[66]:
```
df=pd.DataFrame(df)
plt.pie(x=df['sales'],labels=df['name'],autopct='%1.1f%%')
plt.title('Name Vs Sales')
plt.show()
```




[ ]: